

Yuri Ajala da Costa

Campinas, SP, Brazil

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Portfolio: linktr.ee/yuri.ajala

Hard Skills & Core Stack

Programming & Tools: Python, MATLAB, R, C#, SQL, Git, Linux/WSL2

Core Domains:
Machine Learning, Signal & Image Processing, Quantitative MRI / fMRI, R&D, Data Modeling, Biostatistics

Engineering & Processes:
Product Development, Standard Operating Procedures (SOP), Technical Documentation

Languages
Portuguese (Native)
English (Advanced/C1)
Spanish (Intermediate)

OBJECTIVE

R&D Engineer | AI/ML Engineer | Data Scientist
Multidisciplinary Electrical Engineer and Data Scientist bridging advanced academic research with high-impact industrial research and development.

PROFESSIONAL EXPERIENCE

UNICAMP — PhD Researcher
Jul 2024 – Present | Campinas, SP, Brazil (Hybrid)

- Building quantitative data pipelines using Python and MATLAB to process complex physiological datasets and map brain fluid dynamics (fMRI/MRI).
- Managing end-to-end data modeling workflows, linear/non-linear regressions, and biostatistics protocols to support clinical neuroscience studies.

Upwork — Data Analyst & Simulation Engineer
Jan 2018 – Present | Campinas, SP, Brazil (Remote)

- Developed custom mathematical models, automated data scraping pipelines, and simulation frameworks for global clients using MATLAB/Simulink and Python, earning a 5.0/5.0 client satisfaction rating.

SporeData Inc. — Junior Data Scientist
Mar 2022 – Nov 2022 | Campo Grande, MS, Brazil (Remote)

- Designed statistical analytics workflows and validated automated machine learning algorithms across diverse industrial datasets.

PROFESSIONAL EXPERIENCE (CONT.)

Fine Instrument Technology (FIT) — Applications Engineer & Analyst

Jan 2019 – Jan 2021 | São Carlos, SP, Brazil (Hybrid)

- Led cross-functional alignment across software, mechanical, and research teams to build portable, time-domain NMR quality control products from scratch.
- Engineered and deployed non-destructive MRI/NMR predictive protocols for the food and agricultural sectors (e.g., meat tenderness, solid fat content) using machine learning techniques.
- Authored core product documentation, user manuals, and Standard Operating Procedures (SOPs).

Escola de Engenharia de São Carlos (USP) — Teaching Assistant (PAE Fellow)

Feb 2018 – Dec 2018 | São Carlos, SP, Brazil (On-site)

- Mentored undergraduate engineering students in the "Embedded Electronic Systems Programming Lab" (Intel 8085 and Arduino) and guided technical reporting.

EDUCATION

- **Doctor of Philosophy (PhD) in Medical Pathophysiology & Neuroimaging**

Universidade Estadual de Campinas (UNICAMP) | Aug 2024 – Jul 2028 (Expected)

  - Focus: Non-invasive brain fluid dynamics, fMRI, and quantitative MRI data pipelines. CAPES scholarship recipient.

- **Master of Business Administration (MBA) in Data Science & Analytics**

Universidade de São Paulo (USP) / ESALQ | Aug 2023 – Jul 2025

  - Capstone Project: Applied non-destructive data modeling using binomial logistic regression on biological datasets.

- **Master of Science (M.S.) in Electrical Engineering (Signal Processing)**

Universidade de São Paulo (USP - EESC/IFSC) | 2016 – 2018

  - Thesis: Developed a model-based machine learning algorithm for full-range liver fat fraction estimation using MRI and open-source GUI tools (Python/MATLAB). CAPES scholar.

- **Bachelor of Science (B.Sc.) in Electrical Engineering**

Universidade Federal de Mato Grosso do Sul (UFMS) | 2008 – 2014

  - Sandwich Year: Western Michigan University (WMU), USA – Sunseeker Solar Car Electronics Team (Battery cooling systems). Science Without Borders scholar.

SELECTED HONORS & AWARDS

- **Best Oral Presentation Award – ISMRM Brazilian Chapter (2025)**
  - Recognized for pioneering research on non-invasive functional MRI and CSF dynamics.
- **Science Without Borders National Delegate & Invited Speaker – CAPES (2014)**
  - Selected among top 40 nationwide scholars; delivered an invited keynote on technological innovation and active learning frameworks.

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